

Problem-Solution Fit Template

Date	15 March 2026
Team ID	Batch 2023-27 ACOE
Project Name	Rising Waters – ML-Based Flood Prediction System
Maximum Marks	3 Marks

Problem-Solution Fit Analysis

S.No	Problem Identified	Solution Feature	Fit Level
1	No real-time ML-powered flood prediction tool available	XGBoost classifier (floods.save) achieving 96.55% accuracy	Strong
2	Complex meteorological data difficult to interpret manually	6-field standardized input form with field descriptions	Strong
3	No emergency response guidance for prediction outputs	Flood Result page with 6 emergency protocol action cards	Strong
4	No accessible web interface for non-technical users	Flask web app with intuitive UI accessible on any browser	Strong
5	No prediction history or audit trail	predictions_log.csv + History page with full timestamped log	Strong
6	Lack of 24/7 cloud accessibility for disaster teams	IBM Cloud Foundry deployment with Docker containerization	Strong
7	Manual comparison of multiple ML models is time-consuming	Home Dashboard with built-in accuracy comparison	Medium